SCSD2613
System Analysis and Design

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

PART IV
The Analysis Process #3:
Structure Chart & Entity Relationship Diagram

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update: August 2019 (sharinhh)

OBJECTIVES

1. Understand the purpose of structure chart.
2. Able to transform DFD to structure chart
3. Understand the basic ERD

MAJOR TOPICS

STRUCTURE CHART	• Structure chart symbols • Guideline for drawing structure chart • Translating DFD to Structure Chart
ENTITY RELATIONSHIPS DIAGRAM (ERD)	• ERD basic concepts • DFD and ERD

OBJECTIVES

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MAJOR TOPICS

STRUCTURE CHART

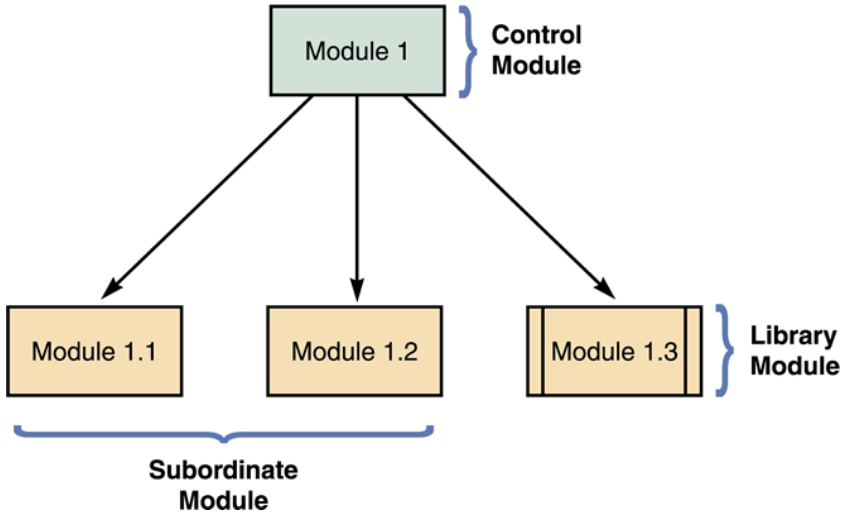
- Structure chart symbols
- Guideline for drawing structure chart
- Translating DFD to Structure Chart

STRUCTURE CHART

- A top-down representation of business functions and processes.
- Also called a functional decomposition diagram.
- Show the program modules and the relationships among them.

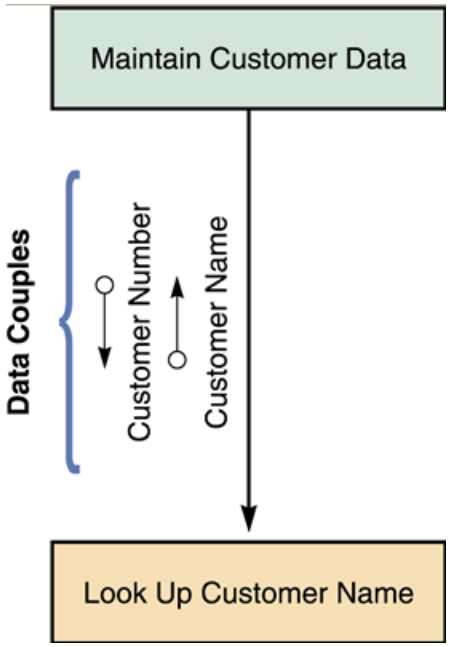
STRUCTURE CHART SYMBOLS

- Module type:
  - **CONTROL MODULE** - a higher-level module that directs lower-level modules
  - **SUBORDINATE MODULES** – a lower-level module in a structure chart.
  - **LIBRARY MODULE** - a module that is reusable and can be invoked from more than one point in the chart.



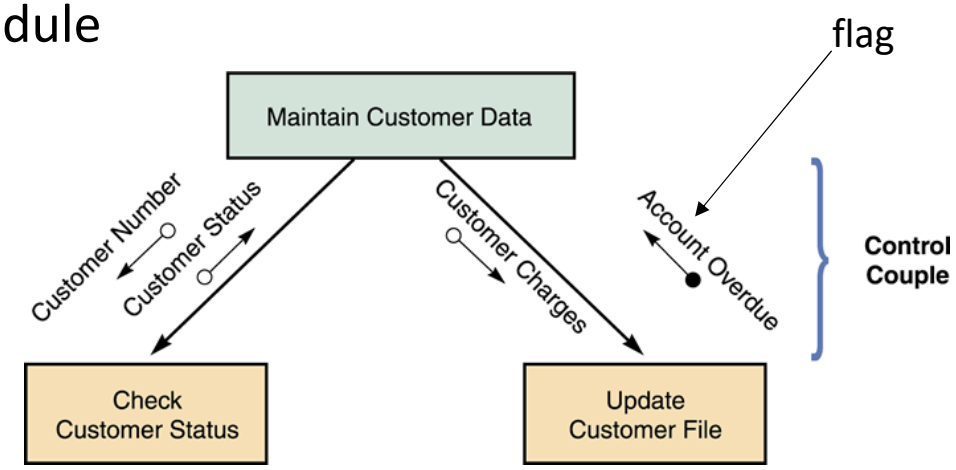
STRUCTURE CHART SYMBOLS

- **DATA COUPLE** - shows data that one module passes to another.



- **CONTROL COUPLE-** shows a message (flag) which one module sends to another.

- **FLAG** - is a message sent from one module to another. A module uses a flag to signal a specific condition or action to another module

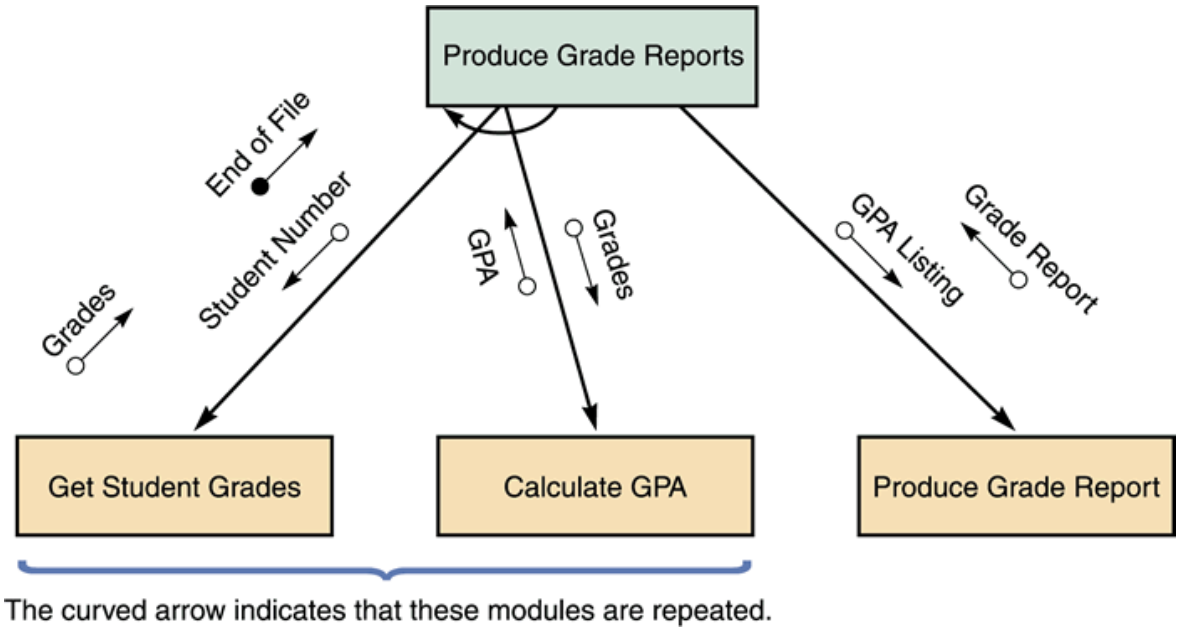


STRUCTURE CHART SYMBOLS

- **CONDITION**
  - A specified action or state in a structure chart.
  - A condition line indicates that a control module determines which subordinate modules will be invoked, depending on a specific condition

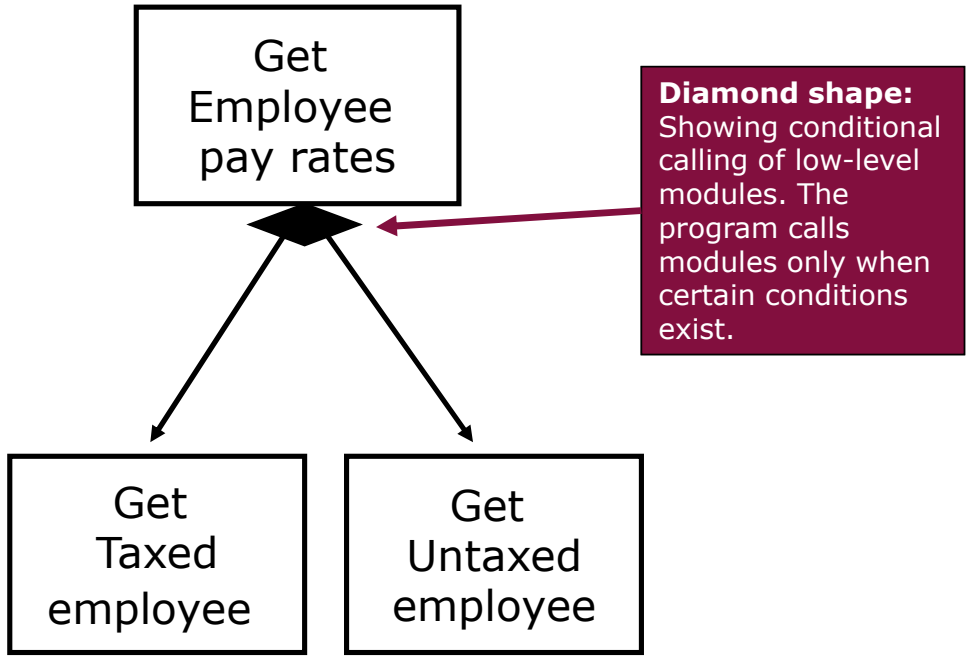
- **LOOP**
  - indicates that one or more modules are repeated

An example of structure chart loop



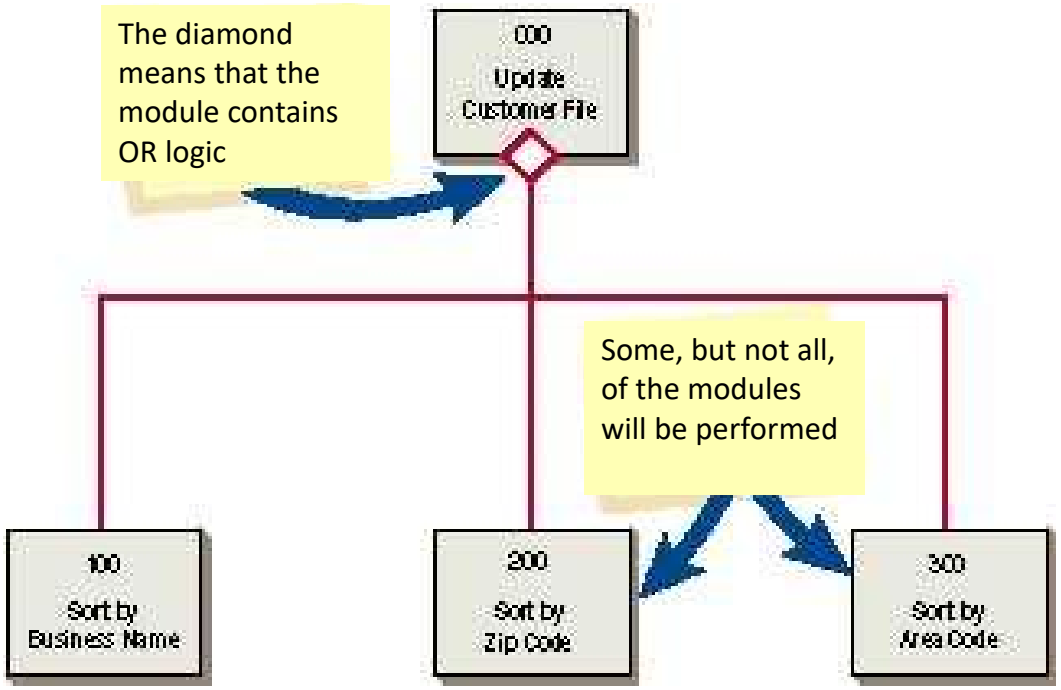
STRUCTURE CHART – OR Logic

```
START Module Get Employee pay rates
If employee >= 3000
    …
Else if employee <3000
    …
END if
END START
```



STRUCTURE CHART – OR Logic

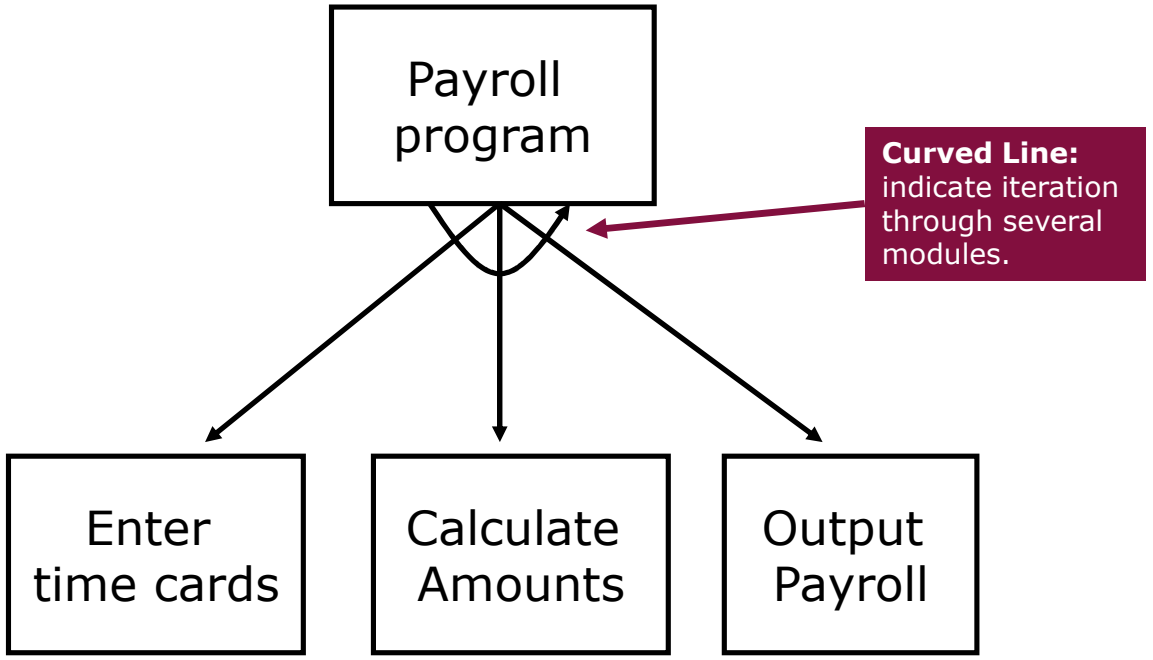
Example : Prepare Customer Bill




```
Start Module 000
If  update = BusName
    Do module 100
Elseif  update = ZipCode
    Do module 200
Else
    Do module 300
End Module A
```

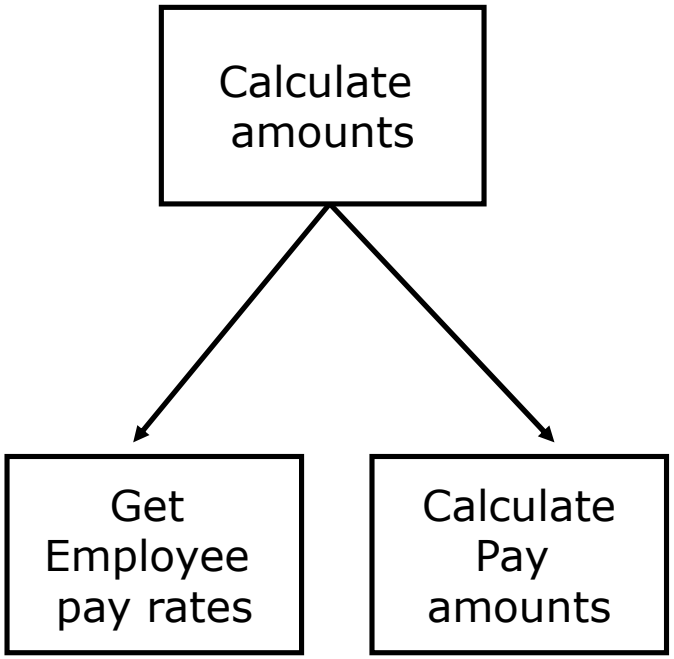
STRUCTURE CHART – WHILE Logic

```
START Program
DoUntil No more time cards
    Call Enter time cards
    Call Calculate amounts
    Call Output payroll
END Until
END START
```



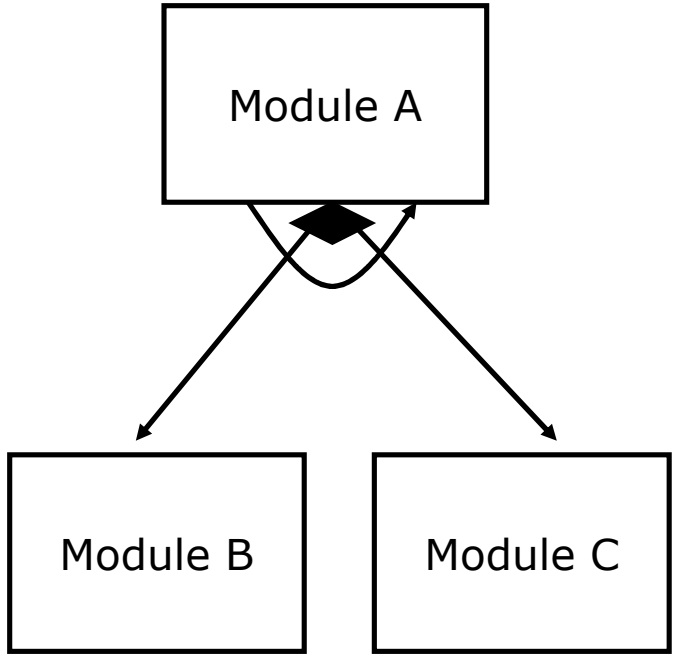
STRUCTURE CHART – without OR & WHILE Logic

```
START Module Calculate amounts
  Call Get employee pay rates
  Call Calculate pay amounts
END START
```



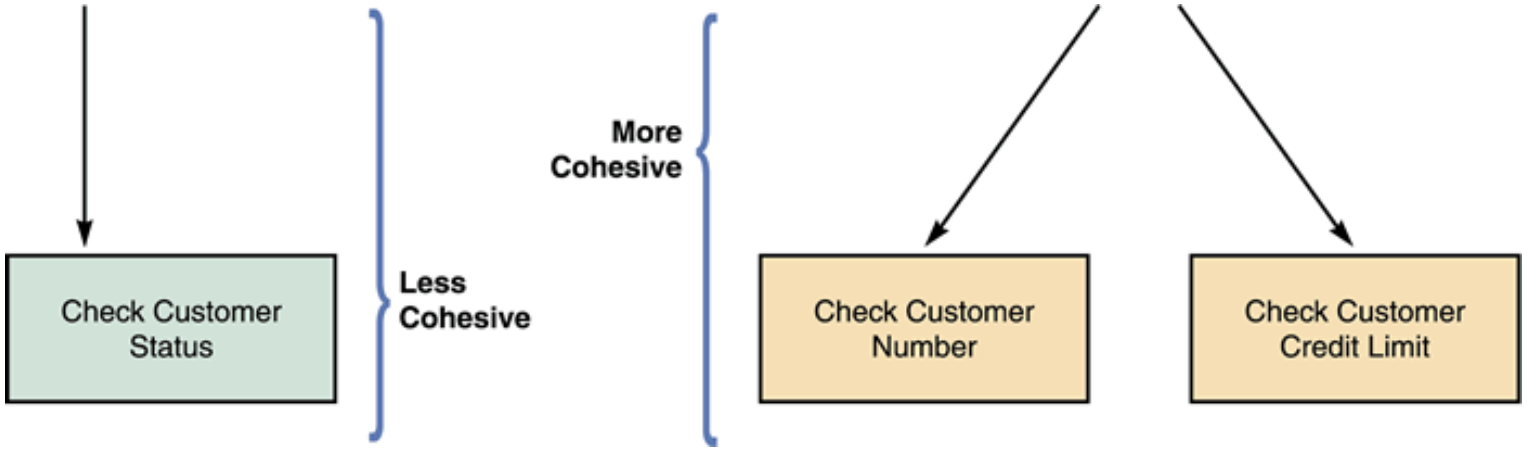
STRUCTURE CHART – with OR & WHILE Logic

```
START Module A
Repeat
If….
 Do Module B
Else…
 Do Module C
END Repeat
END START
```



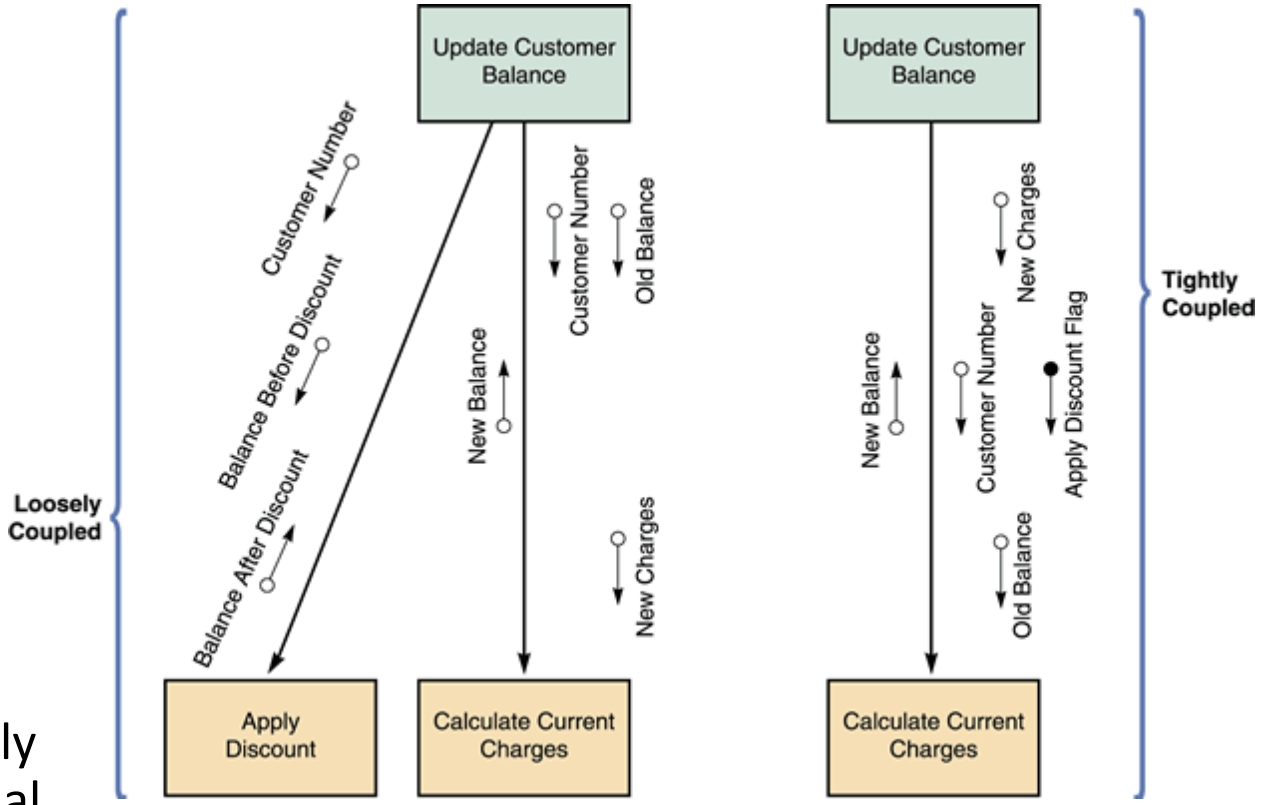
STRUCTURE CHART - COHESION

- Measures a module's scope and processing characteristics.
- A module that performs a single function or task has a high degree of cohesion, which is desirable.



STRUCTURE CHART - COUPLING

- A specified action or state in a structure chart.
- If you need to make a module more cohesive, you can split it into separate units, each of which performs a single function
  - Loosely coupled : Modules that are relatively independent. Loosely coupled modules are easier to maintain and modify, because the logic in one module does not affect other modules.
  - Tightly coupled : If modules are tightly coupled, one module refers to internal logic contained in another module.



GUIDELINE FOR DRAWING A STRUCTURE CHART

STEP 1

Review the DFDs

- Review all DFDs for accuracy and completeness

STEP 2

Identify Modules and Relationships

- Transform functional primitives or object methods into program modules

STEP 3

Add Couples, Loops, and Conditions

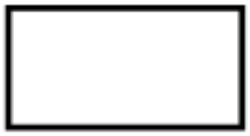
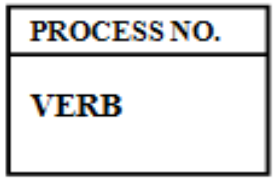
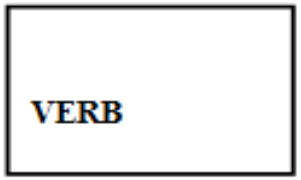
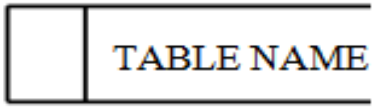
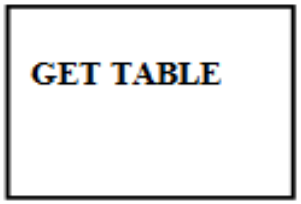
- Identify the data elements that pass from one module to another

STEP 4

Analyze the Structure Chart and the Data Dictionary

- Ensure that the chart reflects all previous documentation and that the logic is correct

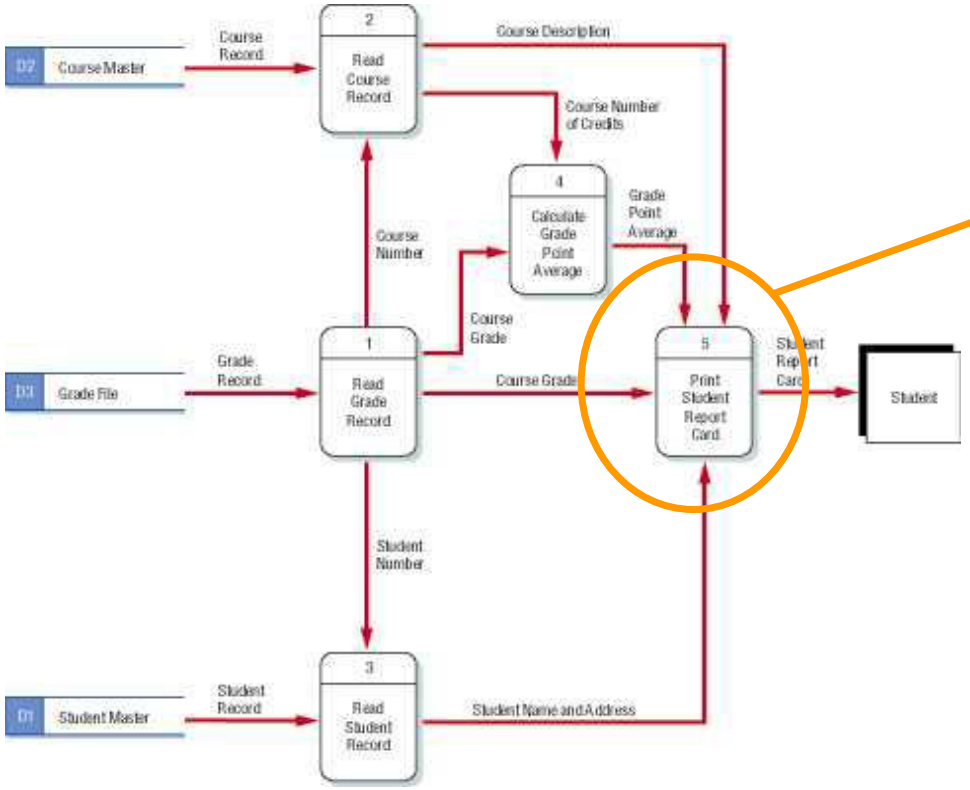
TRANSLATING DFD TO STUCTURE CHART

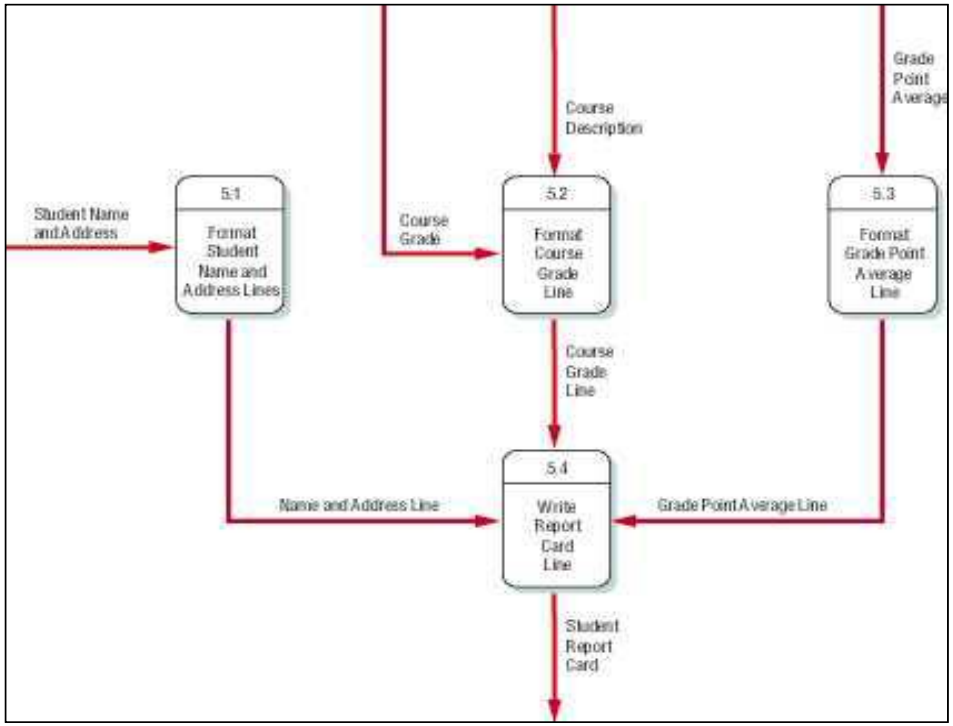
ELEMENTS	DFD	S/CHART
ENTITIES		NONE
PROCESS		
FILE		

TRANSLATING DFD TO STUCTURE CHART

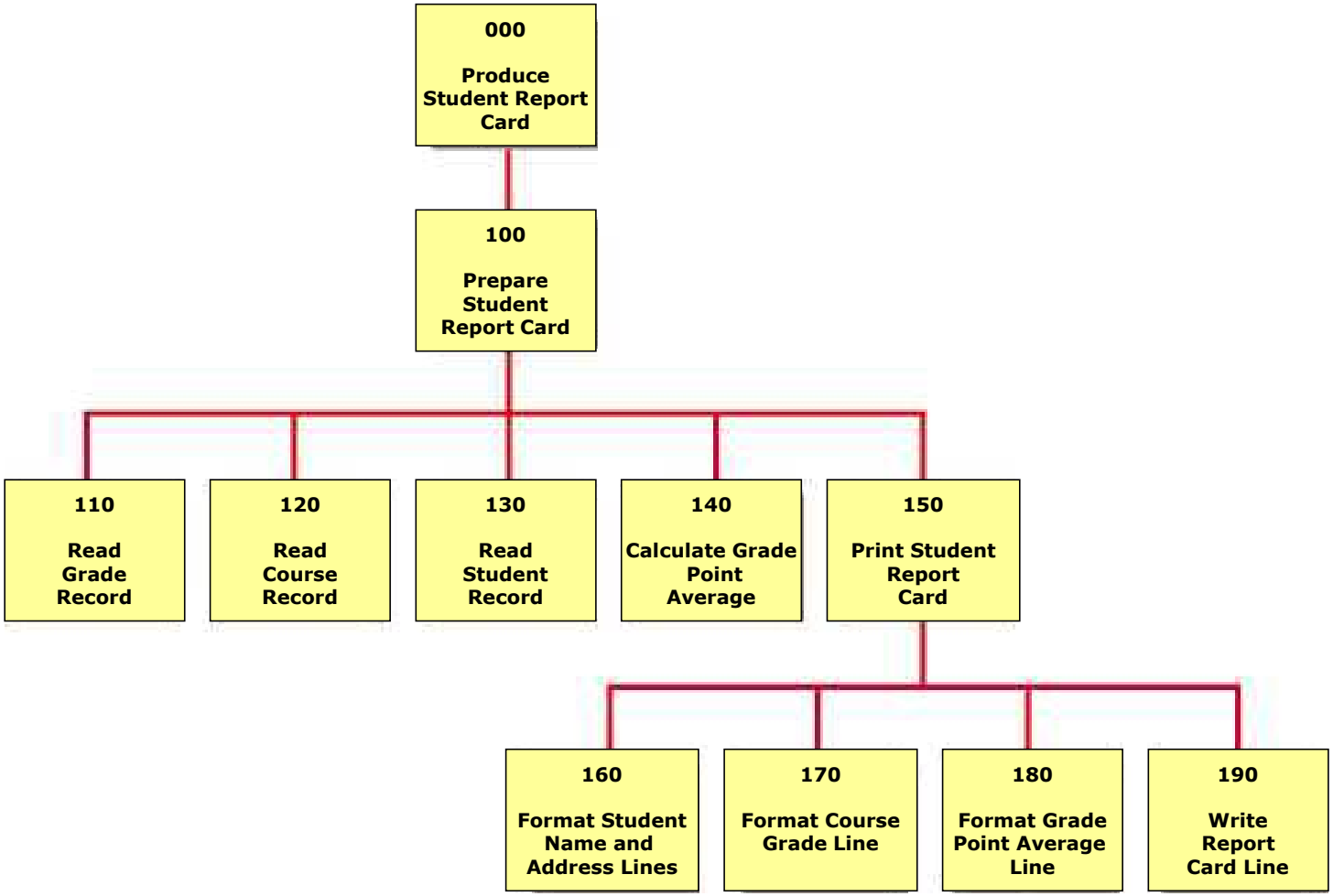
ELEMENTS	DFD	S/CHART
DATA FLOW	 DATA FLOW NAME	 DATA FLOW NAME
FLAG	NONE	 FLAG NAME/LABEL
CALL	NONE	

FROM DFD TO STUCTURE CHART

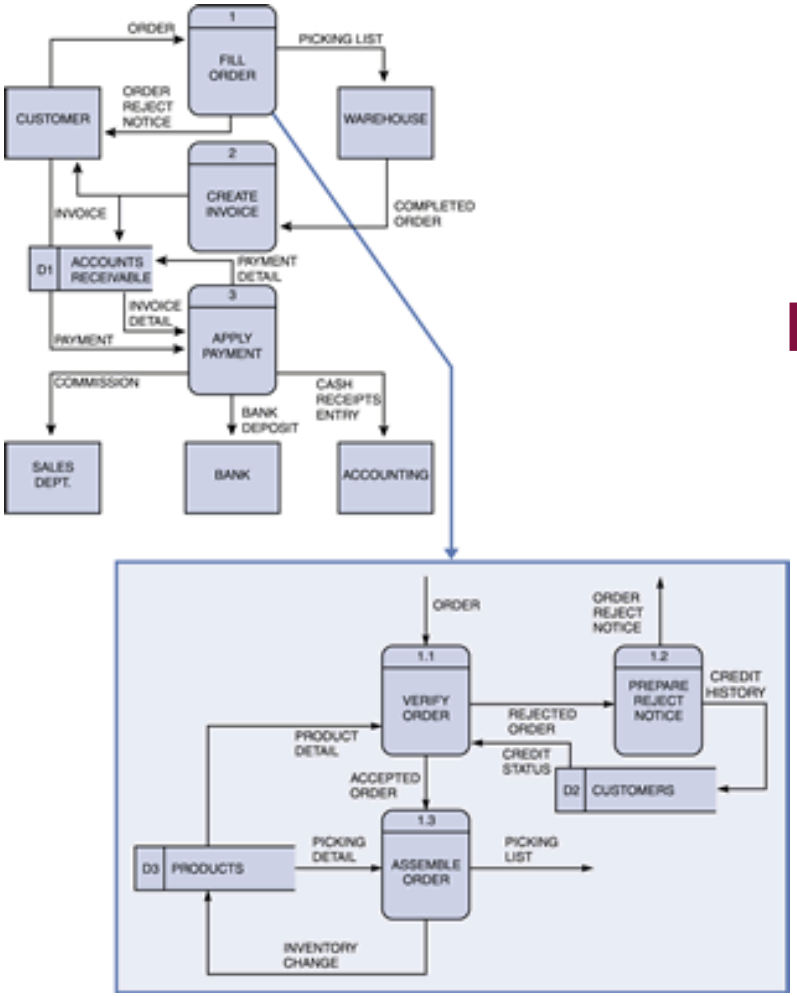


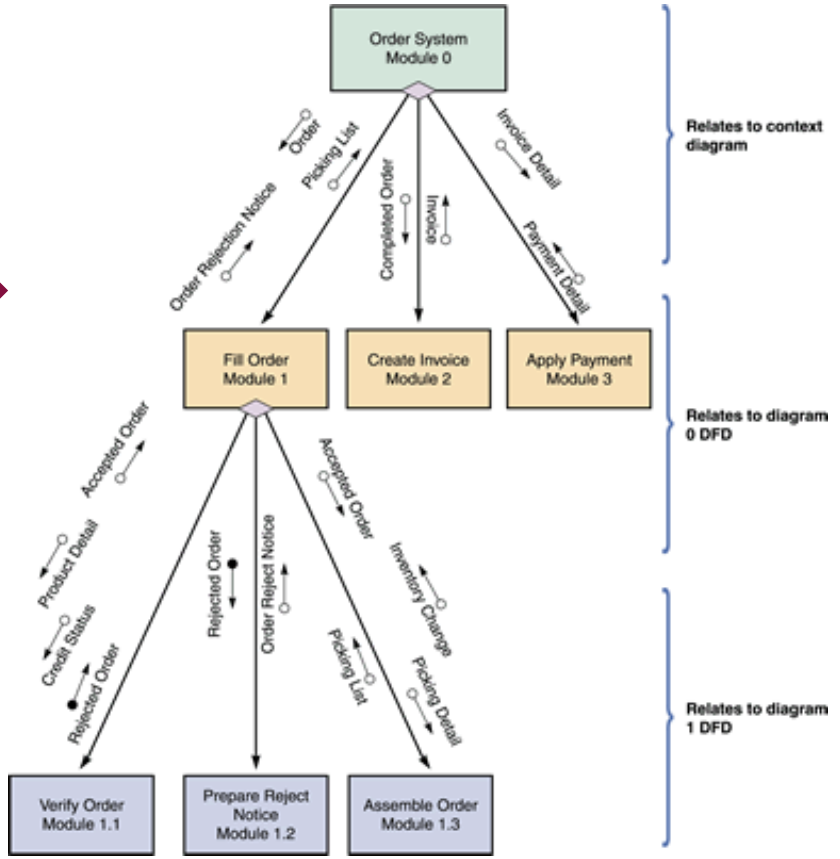


FROM DFD TO STUCTURE CHART



DFD TO STUCTURE CHART Example





LET'S TRY

Checkpoint 1: DFD to Structure Chart (Final 2016/2017)

Figure 1 shows the physical Data Flow Diagram for Book Rental System. Based on Figure 1, convert the DFD into a structure chart.

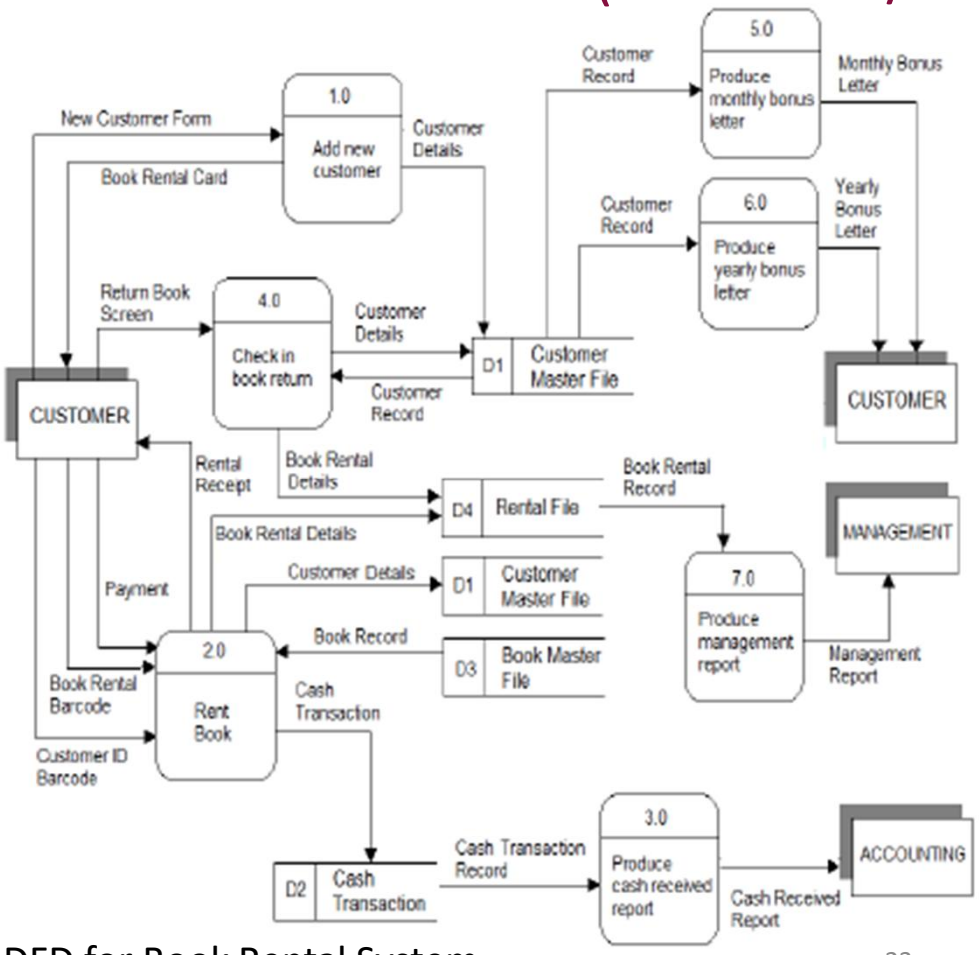


Figure 1: Physical DFD for Book Rental System

MAJOR TOPICS

ENTITY RELATIONSHIPS DIAGRAM

- ERD basic concepts
- DFD and ERD

WHAT IS ER DIAGRAM?

- A diagram used to define the data storage requirements is called the entity-relationship diagram (ERD)
- Data storage requirements include the data entities, their attributes, and the relationships among the data entities.

ERD BASIC CONCEPTS

CONCEPTS	DESCRIPTIONS
Entity type	Group of objects with same properties, identified by enterprise as having an independent existence. A class of persons, places, objects, events, or concepts about which we need to capture and store data. Examples are person, object, place, event, and concepts.
Entity occurrence	Uniquely identifiable object of an entity type.
Relationship type	Set of meaningful associations among entity types.
Relationship occurrence	Uniquely identifiable association, which includes one occurrence from each participating entity type.
Degree of a Relationship	Number of participating entities in relationship. It is limited by constraints such as cardinality and participation (mandatory {All}, optional {Not All}, disjoint {or}), nondisjoint {and})
Relationship of degree	A degree of relationship, for examples are : binary, ternary, quaternary or n-nary relationship

ERD BASIC CONCEPTS

CONCEPTS	DESCRIPTIONS
Attributes	A descriptive property or characteristics of an entity. Synonyms include element, property and field. In example as Student (Entity), StudentID (Attribute). There are many type of attributes such as simple, composite, single-valued, multi-valued and derived attribute.
Keys	An attribute, or a group of attributes, that assumes a unique value for each entity instance. It is sometimes called an identifier. There are candidate key, primary key, alternate key, foreign key and composite key.
Generalization	A concept wherein the attributes that are common to several types of an entity (inheritance relationship)
Supertype	An entity whose instances store attributes that are common to one or more entity substypes.
Subtype	An entity whose instances may inherit common attributes from its entity supertype.

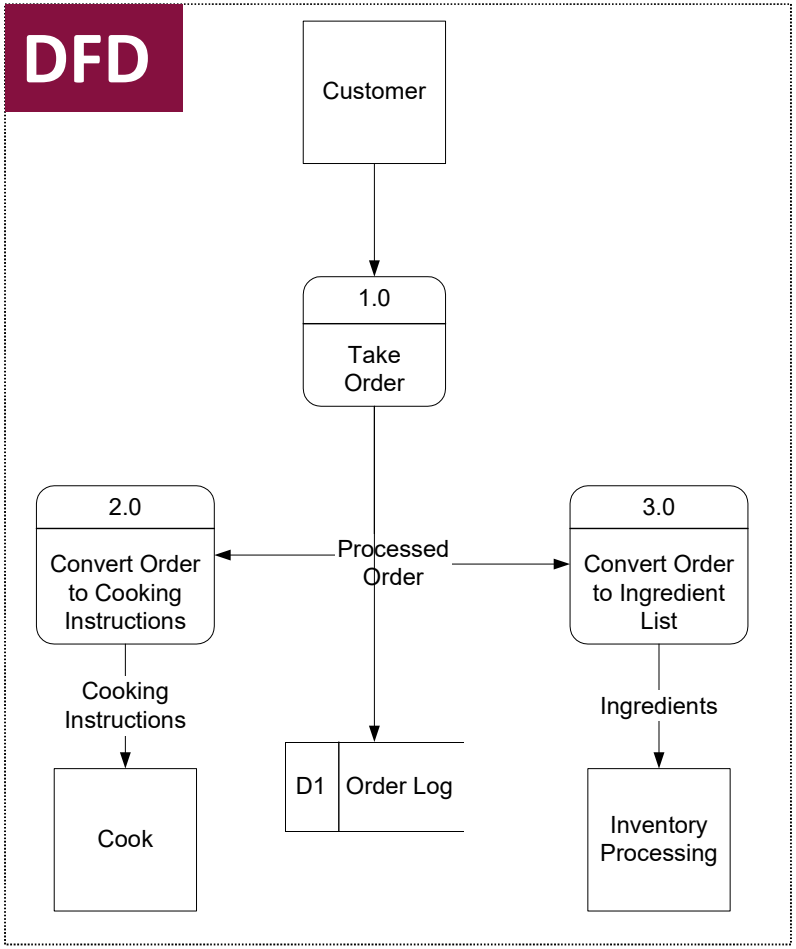
DATA FLOW DIAGRAM (DFD) AND ENTITY RELATIONSHIP DIAGRAM (ERD)

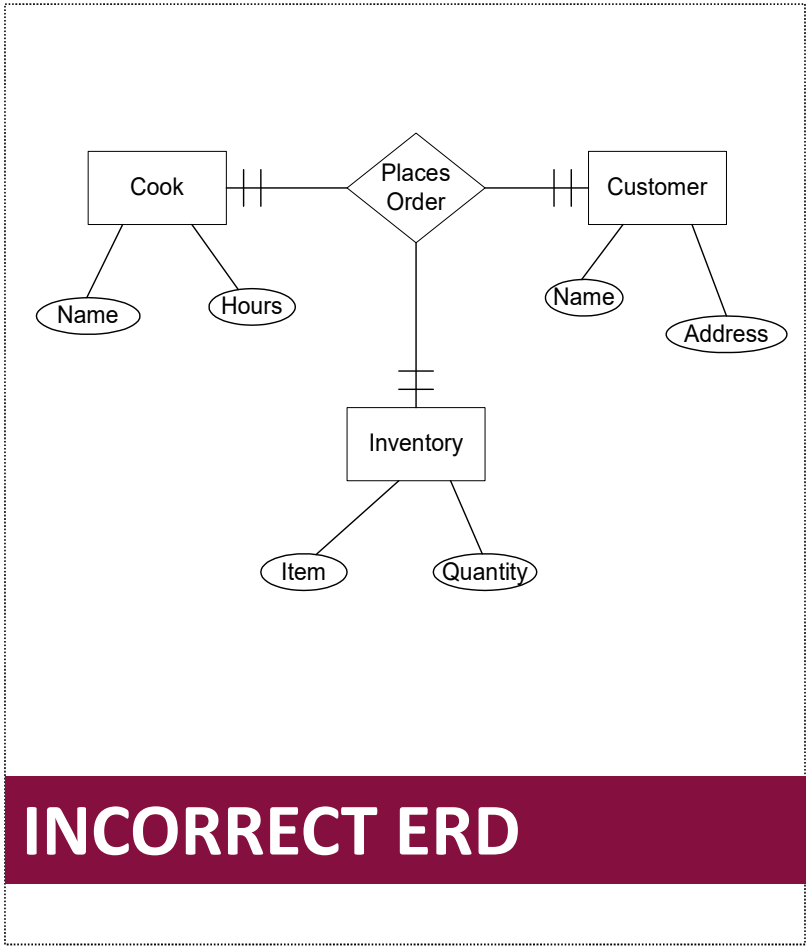
- DFDs and ERDs are both used to model systems, but they show two very different perspectives on the system
- A DFD shows what the system ***does*** as well as the ***data*** that the system manipulates
- An ERD shows **only** the ***data*** that the system manipulates.

DATA FLOW DIAGRAM (DFD) AND ENTITY RELATIONSHIP DIAGRAM (ERD)

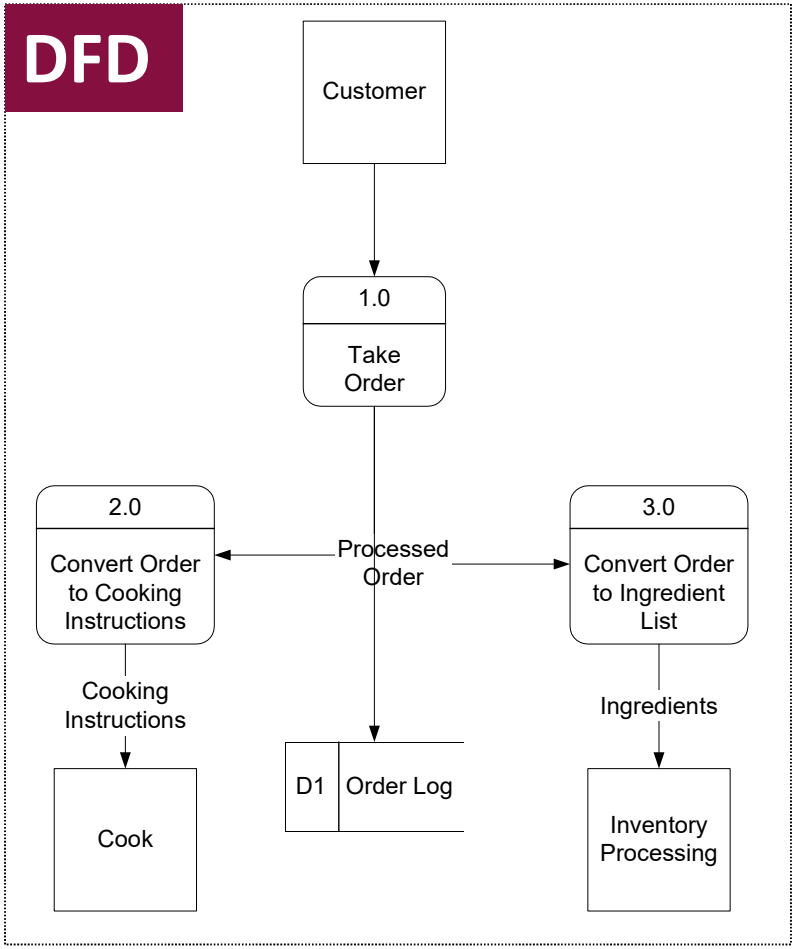
- **ENTITIES** on an ERD often (but not always) correspond to **DATA STORES** on a DFD
- **ATTRIBUTES** on an ERD usually correspond to **DATA ELEMENTS** (listed in the data dictionary) that make up the data store and data flows on a DFD
- Relationships on an ERD **DO NOT** correspond to processes on a DFD.
- Sources on a DFD usually **DO NOT** show up as entities on an ERD

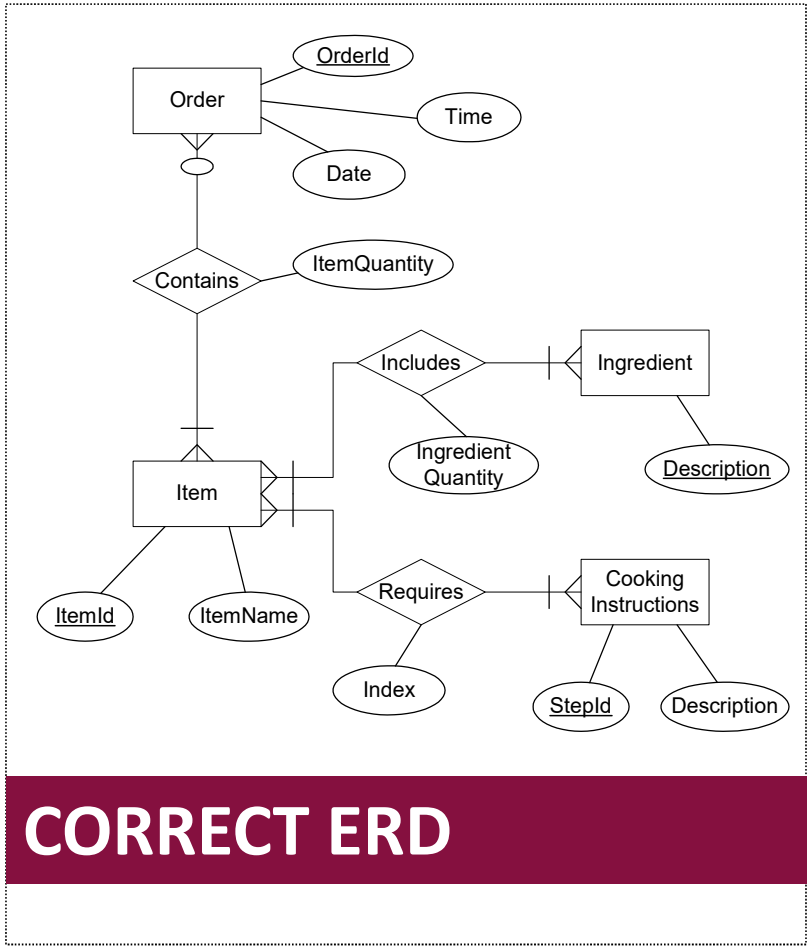
EXAMPLE OF DFD AND ERD



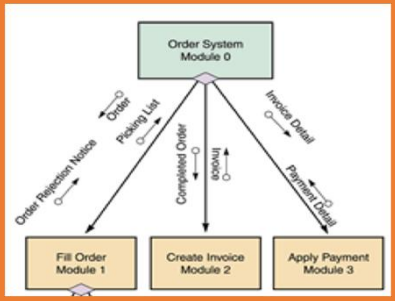


EXAMPLE OF DFD AND ERD



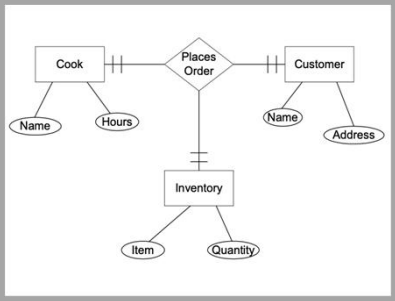


WHEN SHOULD BE USED?



STRUCTURE CHART

When we want to design a program code, usually we will **transform logical DFD** (to-be) to a set of structure chart. From a structure chart, we can write a pseudocode or draw a flow chart before to start programming.



ER DIAGRAM

We will need this very useful data modeling tool when to design database. **From DFD**, we have many data storage elements which will may to be automated. We use E-R diagram to design database for them. Meaning, how to systematically store all data as computer format

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Thank You

update: August 2019 (sharinhh)

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